

ELITE IGCSE ACADEMY

Private Past Paper Solutions

Jan 2021 P2H Worked Solutions

Jan 2021 | P2H

25 questions ■ question image followed by worked solution

PREPARED BY

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PAST PAPER QUESTION**Q#: 1****Marks: 3****TOPIC: Standard & Compound Units**

Jan 2021 P2H - Jan 2021 | P2H

- 1** A train takes 6 hours 39 minutes to travel from New Delhi to Kanpur.
The train travels a distance of 429 km.

Work out the average speed of the train.

Give your answer in km/h correct to one decimal place.

..... km/h

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION**Q#: 1** **Marks: 3****TOPIC: Standard & Compound Units**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected to Standard & Compound Units.**Solution**

The time is

$$6 \text{ h } 39 \text{ min} = 6.65 \text{ h}$$

Average speed:

$$429 \div 6.65 = 64.511 \dots$$

FINAL ANSWER

64.5 km/h.

PAST PAPER QUESTION**Q#: 2****Marks: 3**TOPIC: **Statistics Toolkit**

Jan 2021 P2H - Jan 2021 | P2H

- 2 Ava writes down five whole numbers.

For these five numbers

the median is 7

the mode is 8

the range is 5

Find a possible value for each of the five numbers that Ava writes down.

.....

(Total for Question 2 is 3 marks)

PAST PAPER SOLUTION**Q#: 2****Marks: 3**TOPIC: **Statistics Toolkit**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Statistics Toolkit. The tag is correct.**Solution**

The median is 7, so the middle number is 7.

The mode is 8, so use two 8s.

The range is 5, so choose the smallest and largest numbers to differ by 5.

One possible set is

3, 6, 7, 8, 8

Range:

$$8 - 3 = 5$$

Mode:

8

Median:

7

FINAL ANSWER

3, 6, 7, 8, 8.

PAST PAPER QUESTION**Q#: 3****Marks: 6****TOPIC: Percentages**

Jan 2021 P2H - Jan 2021 | P2H

- 3 Gladys buys a table for \$465 to sell in her shop.

She sells the table for \$520

- (a) Work out the percentage profit that Gladys makes from the sale of the table.
Give your answer correct to 3 significant figures.

..... %
(3)

Gladys has a sale in her shop.

She decreases all the normal prices by 12%
The normal price of an armchair was \$550

- (b) Work out the sale price of the armchair.

\$.....
(3)

(Total for Question 3 is 6 marks)

PAST PAPER SOLUTION**Q#: 3****Marks: 6**TOPIC: **Percentages**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Percentages. The tag is correct.**Method**

Profit:

$$520 - 465 = 55$$

Percentage profit:

$$\frac{55}{465} \times 100 = 11.827 \dots$$

Correct to 3 significant figures:

$$11.8\%$$

For the sale price, a 12% decrease means 88% of the original price:

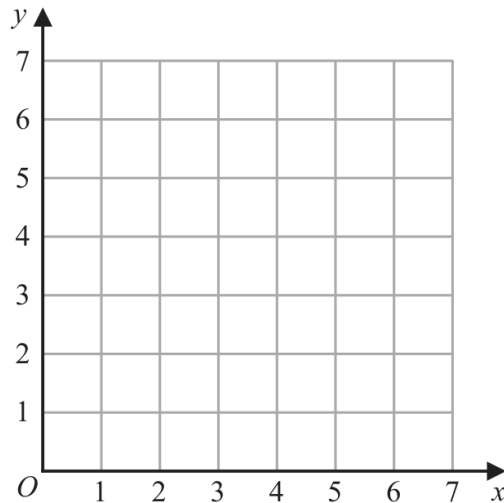
$$550 \times 0.88 = 484$$

FINAL ANSWER

(a) 11.8%, (b) \$484

PAST PAPER QUESTION**Q#: 4****Marks: 4****TOPIC: Graphing Inequalities**

Jan 2021 P2H - Jan 2021 | P2H

4(a) On the grid, draw and **label** the straight line with equation

(i) $x = 1.5$

(ii) $y = x$

(iii) $x + y = 6$

(3)(b) Show, by shading on the grid, the region that satisfies **all three** of the inequalities

$x \geq 1.5$

$y \geq x$

$x + y \leq 6$

Label the region **R**.**(1)****(Total for Question 4 is 4 marks)**

PAST PAPER SOLUTION**Q#: 4****Marks: 4****TOPIC: Graphing Inequalities**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct. This is drawing and shading inequalities.**Solution**

Draw the three boundary lines:

$$x = 1.5, \quad y = x, \quad x + y = 6$$

The required region satisfies

$$x > 1.5$$

$$y \geq x$$

$$x + y \leq 6$$

FINAL ANSWERShade the region $x > 1.5$, $y \geq x$, $x + y \leq 6$.

PAST PAPER QUESTION**Q#: 5****Marks: 7****TOPIC: Solving Inequalities**

Jan 2021 P2H - Jan 2021 | P2H

- 5 (a) Expand and simplify $4x(2x + 5) - 3x(2x - 3)$

.....
(2)

Given that $\frac{y^5 \times y^n}{y^6} = y^{13}$

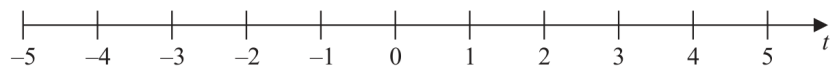
- (b) work out the value of n .

$n =$
(2)

- (c) (i) Solve the inequality $7t - 8 < 2t + 7$

.....
(2)

- (ii) On the number line below, represent the solution set of the inequality solved in part (c)(i)



(1)

(Total for Question 5 is 7 marks)

PAST PAPER SOLUTION**Q#: 5****Marks: 7****TOPIC: Solving Inequalities**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected to solving inequalities. The original tag was expanding brackets, but the inequality and number line form the main single part.

Solution

(a)

$$4x(2x + 5) - 3x(2x - 3) = 8x^2 + 20x - 6x^2 + 9x = 2x^2 + 29x$$

(b)

$$\frac{y^5 \times y^n}{y^6} = y^{n-1} = y^{13}$$

so $n - 1 = 13$, hence $n = 14$.

(c)

$$7t - 8 < 2t + 7$$

$$5t < 15$$

$$t < 3$$

On the number line, use an open circle at 3 and shade to the left.

FINAL ANSWER

(a) $2x^2 + 29x$, (b) $n = 14$, (c) $t < 3$

PAST PAPER QUESTION**Q#: 6****Marks: 4****TOPIC: Powers, Roots & Standard Form**

Jan 2021 P2H - Jan 2021 | P2H

- 6 (a) Write down the value of y^0

.....
(1)

(b) Work out $\frac{9.6 \times 10^{141} + 6.4 \times 10^{140}}{3.2 \times 10^{16}}$

Give your answer in standard form.

.....
(3)

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION**Q#: 6****Marks: 4****TOPIC: Powers, Roots & Standard Form**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Powers, roots and standard form. The tag is correct.**Method**

Any non-zero value to the power 0 is 1, so

$$y^0 = 1$$

For part (b),

$$9.6 \times 10^{141} + 6.4 \times 10^{140} = 9.6 \times 10^{141} + 0.64 \times 10^{141}$$

$$= 10.24 \times 10^{141} = 1.024 \times 10^{142}$$

Now divide:

$$\frac{1.024 \times 10^{142}}{3.2 \times 10^{16}} = 0.32 \times 10^{126}$$

$$= 3.2 \times 10^{125}$$

FINAL ANSWER(a) 1, (b) 3.2×10^{125}

PAST PAPER QUESTION**Q#: 7****Marks: 3**TOPIC: **Statistics Toolkit**

Jan 2021 P2H - Jan 2021 | P2H

- 7 There are 5 cocoa pods in a bag.
The mean weight of the 5 cocoa pods is 398 grams.
- A sixth cocoa pod is put into the bag.
The mean weight of the 6 cocoa pods is 401 grams.
- Work out the weight of the sixth cocoa pod that is put into the bag.

..... grams

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION**Q#: 7****Marks: 3**TOPIC: **Statistics Toolkit**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Statistics Toolkit. The tag is correct.**Solution**

Total weight of 5 cocoa pods:

$$5 \times 398 = 1990 \text{ g}$$

Total weight of 6 cocoa pods:

$$6 \times 401 = 2406 \text{ g}$$

Weight of the sixth cocoa pod:

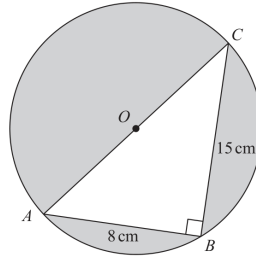
$$2406 - 1990 = 416$$

FINAL ANSWER

416 grams.

PAST PAPER QUESTION**Q#: 8****Marks: 5**TOPIC: **Area & Perimeter**

Jan 2021 P2H - Jan 2021 | P2H

8 A, B and C are points on a circle with centre O .Diagram **NOT**
accurately drawn AOC is a diameter of the circle. $AB = 8 \text{ cm}$ $BC = 15 \text{ cm}$ Angle $ABC = 90^\circ$ Work out the total area of the regions shown shaded in the diagram.
Give your answer correct to 3 significant figures...... cm^2

(Total for Question 8 is 5 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 5**TOPIC: **Area & Perimeter**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Area & Perimeter. The tag is correct.**Solution**Triangle ABC is right-angled:

$$AC^2 = 8^2 + 15^2 = 64 + 225 = 289$$

$$AC = 17 \text{ cm}$$

Since AOC is a diameter, the radius of the circle is

$$17 \div 2 = 8.5 \text{ cm}$$

Area of the circle:

$$\pi(8.5)^2 = 226.98 \dots \text{ cm}^2$$

Area of triangle ABC :

$$\frac{1}{2} \times 8 \times 15 = 60 \text{ cm}^2$$

The shaded area is

$$226.98 \dots - 60 = 166.98 \dots$$

FINAL ANSWER167 cm².

PAST PAPER QUESTION**Q#: 9****Marks: 2****TOPIC: Prime Factors, HCF & LCM**

Jan 2021 P2H - Jan 2021 | P2H

9

$$A = 2^3 \times 3^2 \times 5^2 \times 11$$

$$B = 2^4 \times 3 \times 5^4 \times 13$$

Find the lowest common multiple (LCM) of A and B .
Give your answer as a product of powers of prime numbers.

.....
(Total for Question 9 is 2 marks)

PAST PAPER SOLUTION**Q#: 9****Marks: 2****TOPIC: Prime Factors, HCF & LCM**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Prime factors, HCF and LCM. The tag is correct.**Method**

$$A = 2^3 \times 3^2 \times 5^2 \times 11$$

$$B = 2^4 \times 3 \times 5^4 \times 13$$

For the LCM, take the highest power of every prime that appears:

$$\text{LCM} = 2^4 \times 3^2 \times 5^4 \times 11 \times 13$$

FINAL ANSWER

$$2^4 \times 3^2 \times 5^4 \times 11 \times 13$$

PAST PAPER QUESTION**Q#: 10****Marks: 3**TOPIC: **Percentages**

Jan 2021 P2H - Jan 2021 | P2H

- 10** The people working for a company work in Team A or in Team B.

number of people in Team A : number of people in Team B = 3 : 4

$\frac{4}{5}$ of Team A work full time.

24% of Team B work full time.

Work out what fraction of the people working for the company work full time.
Give your fraction in its simplest form.

.....
(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION**Q#: 10****Marks: 3**TOPIC: **Percentages**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Percentages. The tag is correct.**Method**

The ratio of Team A to Team B is 3 : 4.

Use 3 parts for Team A and 4 parts for Team B.

Full-time part from Team A:

$$\frac{4}{5} \times 3 = \frac{12}{5}$$

Full-time part from Team B:

$$24\% \times 4 = \frac{24}{100} \times 4 = \frac{24}{25}$$

Total full-time parts:

$$\frac{12}{5} + \frac{24}{25} = \frac{60}{25} + \frac{24}{25} = \frac{84}{25}$$

Total parts:

$$3 + 4 = 7$$

So the fraction who work full time is

$$\frac{\frac{84}{25}}{7} = \frac{84}{175} = \frac{12}{25}$$

FINAL ANSWER

$$\frac{12}{25}$$

PAST PAPER QUESTION**Q#: 11****Marks: 3**TOPIC: **Algebraic Roots & Indices**

Jan 2021 P2H - Jan 2021 | P2H

11 Simplify fully $\left(\frac{9t^4w^9}{18t^6w^{10}}\right)^{-2}$

(Total for Question 11 is 3 marks)

Dr. Eslam

PAST PAPER SOLUTION**Q#: 11****Marks: 3****TOPIC: Algebraic Roots & Indices**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected to Algebraic Roots & Indices.**Solution**

$$\left(\frac{9t^4w^9}{18t^6w^{10}} \right)^{-2}$$

Simplify inside the brackets:

$$\frac{9t^4w^9}{18t^6w^{10}} = \frac{1}{2}t^{-2}w^{-1} = \frac{1}{2t^2w}$$

Now apply the power -2 :

$$\begin{aligned} \left(\frac{1}{2t^2w} \right)^{-2} &= (2t^2w)^2 \\ &= 4t^4w^2 \end{aligned}$$

FINAL ANSWER

$$4t^4w^2.$$

PAST PAPER QUESTION**Q#: 12****Marks: 2**TOPIC: **Statistics Toolkit**

Jan 2021 P2H - Jan 2021 | P2H

12 15 people were asked how long, in minutes, they had been waiting for a bus.

Here are the results.

2 3 3 4 5 6 6 8 9 10 11 13 14 15 18

Find the interquartile range of these times.

..... minutes

(Total for Question 12 is 2 marks)

PAST PAPER SOLUTION**Q#: 12****Marks: 2**TOPIC: **Statistics Toolkit**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Statistics Toolkit. The tag is correct.**Solution**

The data is already in order:

 $2, 3, 3, 4, 5, 6, 6, 8, 9, 10, 11, 13, 14, 15, 18$

For 15 values,

$$Q_1 = 4, \quad Q_3 = 13$$

$$\text{IQR} = 13 - 4 = 9$$

FINAL ANSWER

9 minutes.

PAST PAPER QUESTION**Q#: 13****Marks: 3**TOPIC: **Circle Theorems**

Jan 2021 P2H - Jan 2021 | P2H

13 P, Q, R, S and T are points on a circle with centre O .

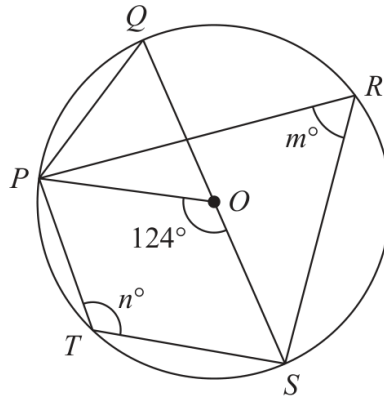


Diagram **NOT**
accurately drawn

QOS is a diameter of the circle.

angle $POS = 124^\circ$ angle $PRS = m^\circ$ angle $PTS = n^\circ$

(a) Find the value of

(i) m

.....

(ii) n

.....

(2)

(b) Find the size of angle QPO .

.....

(1)

(Total for Question 13 is 3 marks)

PAST PAPER SOLUTION**Q#: 13****Marks: 3****TOPIC: Circle Theorems**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected to Circle Theorems. The original tag was simultaneous equations, but this is a circle geometry question.

Solution

The central angle $\angle POS$ is 124° . The angle at the circumference standing on the same chord PS is half this angle.

$$m = \frac{124}{2} = 62$$

Angles opposite each other in a cyclic quadrilateral add to 180° , so

$$n = 180 - 62 = 118$$

Since QOS is a diameter,

$$\angle QOP = 180^\circ - 124^\circ = 56^\circ$$

Triangle QOP is isosceles because $OQ = OP$. Therefore

$$\angle QPO = \frac{180 - 56}{2} = 62^\circ$$

FINAL ANSWER

$m = 62$, $n = 118$, and $\angle QPO = 62^\circ$.

PAST PAPER QUESTION**Q#: 14****Marks: 7****TOPIC: Rearranging Formulas**

Jan 2021 P2H - Jan 2021 | P2H

14 (a) Solve $\frac{9a - 7}{5} - \frac{3a - 7}{4} = 4.55$

Show clear algebraic working.

$$a = \dots\dots\dots$$

(3)

(b) Make c the subject of the formula $p = \sqrt{\frac{ac + 8}{3 + c}}$

$$\dots\dots\dots$$

(4)

(Total for Question 14 is 7 marks)

PAST PAPER SOLUTION**Q#: 14****Marks: 7****TOPIC: Rearranging Formulas**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Rearranging formulas. The tag is correct.**Solution**

(a)

$$\frac{9a - 7}{5} - \frac{3a - 7}{4} = 4.55$$

Multiply by 20:

$$4(9a - 7) - 5(3a - 7) = 91$$

$$36a - 28 - 15a + 35 = 91$$

$$21a + 7 = 91$$

$$a = 4$$

(b)

$$p = \sqrt{\frac{ac + 8}{3 + c}}$$

$$p^2 = \frac{ac + 8}{3 + c}$$

$$p^2(3 + c) = ac + 8$$

$$c(p^2 - a) = 8 - 3p^2$$

FINAL ANSWER

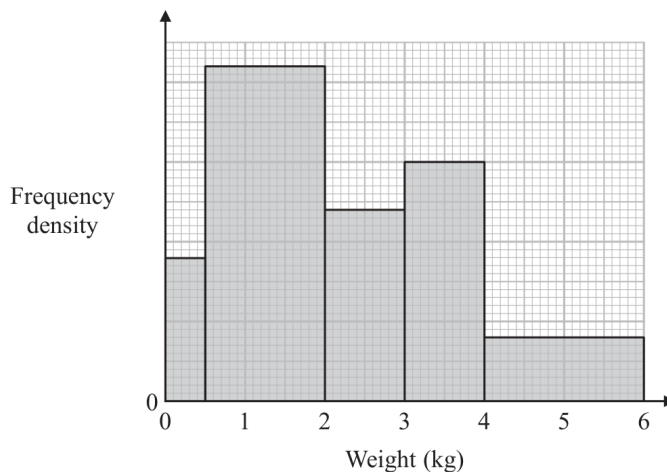
$$(a) a = 4, (b) c = \frac{8 - 3p^2}{p^2 - a}$$

PAST PAPER QUESTION**Q#: 15****Marks: 6****TOPIC: Histograms**

Jan 2021 P2H - Jan 2021 | P2H

15 A postman records the weight of each parcel that he delivers.

The histogram shows information about the weights of all the parcels that the postman delivered last Monday. No parcels weighed more than 6 kg.



63 of the parcels that the postman delivered last Monday each had a weight between 0.5 kg and 2 kg.

(a) Work out the total number of parcels the postman delivered last Monday.

.....
(3)

The postman picks at random two of the records of the parcels he delivered last Monday.

(b) Work out an estimate for the probability that each parcel weighed more than 2.25 kg.

.....
(3)

(Total for Question 15 is 6 marks)

PAST PAPER SOLUTION**Q#: 15****Marks: 6**TOPIC: **Histograms**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Histograms. The tag is correct.**Solution**For $0.5 < w \leq 2$, the frequency is 63, so the density is

$$\frac{63}{1.5} = 42$$

Using the histogram heights, the class frequencies are

$$9, 63, 24, 30, 16$$

So the total number of parcels is

$$9 + 63 + 24 + 30 + 16 = 142$$

For weight greater than 2.25 kg:

$$0.75(24) + 30 + 16 = 64$$

The probability that both randomly chosen records are greater than 2.25 kg is

$$\frac{64}{142} \times \frac{63}{141} = 0.201 \dots$$

FINAL ANSWER

142, then 0.201 approximately.

PAST PAPER QUESTION**Q#: 16****Marks: 6****TOPIC: Set Notation & Venn Diagrams**

Jan 2021 P2H - Jan 2021 | P2H

16 Some students were asked the following question.

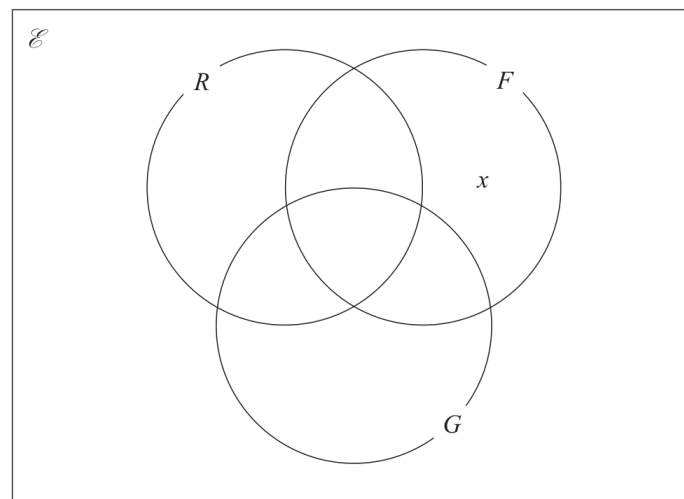
“Which of the subjects Russian (R), French (F) and German (G) do you study?”

Of these students

- 4 study all three of Russian, French and German
- 10 study Russian and French
- 13 study French and German
- 6 study Russian and German
- 24 study German
- 11 study none of the three subjects
- the number who study Russian only is twice the number who study French only.

Let x be the number of students who study French only.

- (a) Show all this information on the Venn diagram, giving the number of students in each appropriate subset, in terms of x where necessary.



(3)

Given that the number of students who were asked the question was 80

- (b) work out the number of these students that study Russian.

(3)

(Total for Question 16 is 6 marks)

PAST PAPER SOLUTION**Q#: 16****Marks: 6****TOPIC: Set Notation & Venn Diagrams**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Set notation and Venn diagrams. The tag is correct.**Method**

The number studying all three languages is 4.

So the two-language-only regions are

$$R \cap F \text{ only} = 10 - 4 = 6$$

$$F \cap G \text{ only} = 13 - 4 = 9$$

$$R \cap G \text{ only} = 6 - 4 = 2$$

The German-only region is

$$24 - 9 - 2 - 4 = 9$$

The French-only region is x and the Russian-only region is $2x$. There are 11 students outside all three sets.

Using the total 80,

$$2x + x + 9 + 6 + 9 + 2 + 4 + 11 = 80$$

$$3x + 41 = 80$$

$$x = 13$$

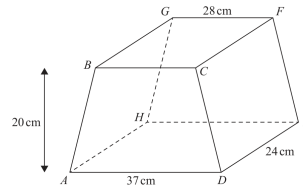
The number studying Russian is

$$2x + 6 + 2 + 4 = 26 + 6 + 2 + 4 = 38$$

FINAL ANSWER**38**

PAST PAPER QUESTIONTOPIC: **Volume & Surface Area**

Jan 2021 P2H - Jan 2021 | P2H

Q#: 17**Marks: 7**17 The diagram shows a solid prism $ABCDEFGH$.Diagram **NOT**
accurately drawn

The trapezium $ABCD$, in which AD is parallel to BC , is a cross section of the prism.
The base $ADEH$ of the prism is a horizontal plane.

$ADEH$ and $BCFG$ are rectangles.

The midpoint of BC is vertically above the midpoint of AD so that $BA = CD$.

$AD = 37\text{ cm}$ $GF = 28\text{ cm}$ $DE = 24\text{ cm}$

The perpendicular distance between edges AD and BC is 20 cm .

(a) Work out the total surface area of the prism.

_____ cm^2
(4)

(b) Calculate the size of the angle between AF and the plane $ADEH$.
Give your answer correct to one decimal place.

_____ $^\circ$
(3)

(Total for Question 17 is 7 marks)

PAST PAPER SOLUTION**Q#: 17****Marks: 7****TOPIC: Volume & Surface Area**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed*

Topic check. Corrected to volume and surface area. The largest part is total surface area of a prism.

Solution

The cross-section is a trapezium with parallel sides 37 cm and 28 cm, and perpendicular height 20 cm.

$$\text{Area of cross-section} = \frac{1}{2}(37 + 28)(20) = 650$$

Since the trapezium is isosceles, the horizontal difference on each side is

$$\frac{37 - 28}{2} = 4.5$$

So

$$AB = CD = \sqrt{20^2 + 4.5^2} = 20.5$$

The perimeter of the cross-section is

$$37 + 28 + 20.5 + 20.5 = 106$$

The length of the prism is 24 cm, so the total surface area is

$$2(650) + 106(24) = 3844$$

For the angle between AF and the horizontal plane, the horizontal projection has length

$$\sqrt{24^2 + 4.5^2}$$

The vertical height is 20, so

$$\tan \theta = \frac{20}{\sqrt{24^2 + 4.5^2}}$$

$$\theta = 39.3^\circ$$

FINAL ANSWER

3844 cm², and 39.3°.

PAST PAPER QUESTION**Q#: 18****Marks: 7****TOPIC: Coordinate Geometry**

Jan 2021 P2H - Jan 2021 | P2H

18 A rectangle $ABCD$ is to be drawn on a centimetre grid such that A has coordinates $(-4, -2)$ B has coordinates $(1, 10)$ C has coordinates $(19, a)$ D has coordinates (b, c) (a) Work out the value of a , the value of b and the value of c . $a =$ _____ $b =$ _____ $c =$ _____

(4)

(b) Calculate the perimeter, in centimetres, of rectangle $ABCD$.

_____ cm

(3)

(Total for Question 18 is 7 marks)

PAST PAPER SOLUTION**Q#: 18****Marks: 7****TOPIC: Coordinate Geometry**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Coordinate geometry. The tag is correct.**Solution**

$$A = (-4, -2), \quad B = (1, 10), \quad C = (19, a), \quad D = (b, c)$$

The vector AB is

$$(1 - (-4), 10 - (-2)) = (5, 12)$$

The vector BC is

$$(19 - 1, a - 10) = (18, a - 10)$$

Adjacent sides of a rectangle are perpendicular, so their dot product is zero:

$$5(18) + 12(a - 10) = 0$$

$$90 + 12a - 120 = 0$$

$$a = \frac{5}{2}$$

Now

$$BC = (18, -7.5)$$

So

$$D = A + BC = (-4, -2) + (18, -7.5) = (14, -9.5)$$

$$b = 14, \quad c = -9.5$$

The side lengths are

$$AB = \sqrt{5^2 + 12^2} = 13$$

$$BC = \sqrt{18^2 + (-7.5)^2} = 19.5$$

Perimeter:

$$2(13 + 19.5) = 65$$

FINAL ANSWER
 $a = \frac{5}{2}$, $b = 14$, $c = -9.5$, and perimeter 65 cm.

PAST PAPER QUESTION**Q#: 19****Marks: 5****TOPIC: Differentiation**

Jan 2021 P2H - Jan 2021 | P2H

19 A particle P is moving along a straight line.The fixed point O lies on this line.At time t seconds where $t \geq 0$, the displacement, s metres, of P from O is given by

$$s = t^3 + 5t^2 - 8t + 10$$

Find the displacement of P from O when P is instantaneously at rest.Give your answer in the form $\frac{a}{b}$ where a and b are integers.

..... metres

(Total for Question 19 is 5 marks)

PAST PAPER SOLUTION**Q#: 19****Marks: 5****TOPIC: Differentiation**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected to differentiation. Instantaneously at rest means the velocity is zero.**Solution**

$$s = t^3 + 5t^2 - 8t + 10$$

Velocity is

$$\frac{ds}{dt} = 3t^2 + 10t - 8$$

At rest,

$$3t^2 + 10t - 8 = 0$$

Factorise:

$$(3t - 2)(t + 4) = 0$$

So

$$t = \frac{2}{3} \quad \text{or} \quad t = -4$$

Since $t \geq 0$,

$$t = \frac{2}{3}$$

Now find s :

$$s = \left(\frac{2}{3}\right)^3 + 5\left(\frac{2}{3}\right)^2 - 8\left(\frac{2}{3}\right) + 10$$

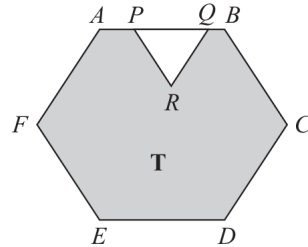
$$s = \frac{8}{27} + \frac{20}{9} - \frac{16}{3} + 10$$

$$s = \frac{194}{27}$$

FINAL ANSWER $\frac{194}{27}$ metres.

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Area & Perimeter**

Jan 2021 P2H - Jan 2021 | P2H

20Diagram **NOT**
accurately drawn

The diagram shows a shaded region **T** formed by removing an equilateral triangle PQR from a regular hexagon $ABCDEF$.

The points P and Q lie on AB such that $AB = 1.5 \times PQ$

Given that the area of region **T** is $72\sqrt{3} \text{ cm}^2$

work out the length of PQ .

.....cm

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Area & Perimeter**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Moved from Angles in Polygons & Parallel Lines to Area & Perimeter.**Method**

Let

$$PQ = p$$

Given

$$AB = 1.5PQ$$

so the side length of the regular hexagon is

$$1.5p = \frac{3}{2}p$$

Area of a regular hexagon with side length s is

$$\frac{3\sqrt{3}}{2}s^2$$

So the area of the hexagon is

$$\begin{aligned} \frac{3\sqrt{3}}{2} \left(\frac{3}{2}p\right)^2 \\ = \frac{27\sqrt{3}}{8}p^2 \end{aligned}$$

The removed triangle PQR is equilateral, so its area is

$$\frac{\sqrt{3}}{4}p^2$$

The shaded area is

$$\begin{aligned} \frac{27\sqrt{3}}{8}p^2 - \frac{\sqrt{3}}{4}p^2 \\ = \frac{25\sqrt{3}}{8}p^2 \end{aligned}$$

This is $72\sqrt{3}$, so

$$\frac{25\sqrt{3}}{8}p^2 = 72\sqrt{3}$$

$$\frac{25}{8}p^2 = 72$$

$$p^2 = \frac{576}{25}$$

$$p = 4.8$$

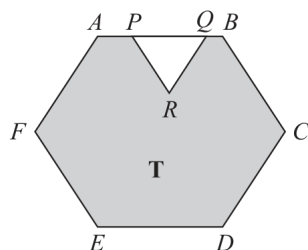
FINAL ANSWER

$$PQ = 4.8 \text{ cm}$$

Dr. Eslam Ahmed

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Area & Perimeter**

Jan 2021 P2H - Jan 2021 | P2H

20Diagram **NOT**
accurately drawn

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The points P and Q lie on AB such that $AB = 1.5 \times PQ$

Given that the area of region **T** is $72\sqrt{3} \text{ cm}^2$

work out the length of PQ .

.....cm

(Total for Question 20 is 4 marks)

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Area & Perimeter**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Moved from Angles in Polygons & Parallel Lines to Area & Perimeter.**Method**

Let

$$PQ = p$$

Given

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so the side length of the regular hexagon is

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Area of a regular hexagon with side length s is

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So the area of the hexagon is

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$$\frac{25\sqrt{3}}{8}p^2 = 72\sqrt{3}$$

$$\frac{25}{8}p^2 = 72$$

$$p^2 = \frac{576}{25}$$

$$p = 4.8$$

FINAL ANSWER

$$PQ = 4.8 \text{ cm}$$

Dr. Eslam Ahmed

PAST PAPER QUESTION**Q#: 21****Marks: 4****TOPIC: Algebraic Fractions**

Jan 2021 P2H - Jan 2021 | P2H

21 Write $\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$

as a single fraction in its simplest form.
Show clear algebraic working.

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 4****TOPIC: Algebraic Fractions**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected from Sine/Cosine Rule to Algebraic Fractions.**Method**

$$\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$$

Factor:

$$25x^2 - 64 = (5x - 8)(5x + 8)$$

$$5x^2 - 13x - 6 = (5x + 2)(x - 3)$$

$$x^2 - 8x + 15 = (x - 3)(x - 5)$$

So

$$\frac{(5x - 8)(5x + 8)}{(5x + 2)(x - 3)} \times \frac{(x - 3)(x - 5)}{5x + 8} - (x - 7)$$

Cancel common factors:

$$\frac{(5x - 8)(x - 5)}{5x + 2} - (x - 7)$$

Use the common denominator $5x + 2$:

$$\frac{(5x - 8)(x - 5) - (x - 7)(5x + 2)}{5x + 2}$$

Expand:

$$(5x - 8)(x - 5) = 5x^2 - 33x + 40$$

$$(x - 7)(5x + 2) = 5x^2 - 33x - 14$$

So the numerator is

$$5x^2 - 33x + 40 - (5x^2 - 33x - 14) = 54$$

FINAL ANSWER

54
$5x + 2$

PAST PAPER QUESTION**Q#: 21****Marks: 4****TOPIC: Algebraic Fractions**

Jan 2021 P2H - Jan 2021 | P2H

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as a single fraction in its simplest form.
Show clear algebraic working.

(Total for Question 21 is 4 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 4****TOPIC: Algebraic Fractions**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Corrected from Sine/Cosine Rule to Algebraic Fractions.**Method**

$$\frac{25x^2 - 64}{5x^2 - 13x - 6} \times \frac{x^2 - 8x + 15}{5x + 8} - (x - 7)$$

Factor:

$$25x^2 - 64 = (5x - 8)(5x + 8)$$

$$5x^2 - 13x - 6 = (5x + 2)(x - 3)$$

$$x^2 - 8x + 15 = (x - 3)(x - 5)$$

So

$$\frac{(5x - 8)(5x + 8)}{(5x + 2)(x - 3)} \times \frac{(x - 3)(x - 5)}{5x + 8} - (x - 7)$$

Cancel common factors:

$$\frac{(5x - 8)(x - 5)}{5x + 2} - (x - 7)$$

Use the common denominator $5x + 2$:

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Expand:

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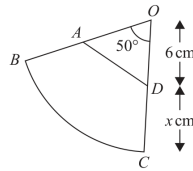
FINAL ANSWER

54
$5x + 2$

PAST PAPER QUESTION**Q#: 22****Marks: 6****TOPIC: Circles, Arcs & Sectors**

Jan 2021 P2H - Jan 2021 | P2H

- 22 The diagram shows a sector $OB C$ of a circle with centre O and radius $(6 + x)$ cm.

Diagram **NOT**
accurately drawn

A is the point on OB and D is the point on OC such that $OA = OD = 6$ cm

Angle $BOC = 50^\circ$

Given that

the perimeter of sector $OB C = 2 \times$ the perimeter of triangle OAD

find the value of x .

Give your answer correct to 3 significant figures.

$x =$

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 6****TOPIC: Circles, Arcs & Sectors**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**The radius of sector OBC is

$$6 + x$$

The angle is 50° , so the arc length of sector OBC is

$$\frac{50}{360} \times 2\pi(6 + x)$$

Therefore the perimeter of sector OBC is

$$\begin{aligned} 2(6 + x) + \frac{50}{360} \times 2\pi(6 + x) \\ = (6 + x) \left(2 + \frac{5\pi}{18} \right) \end{aligned}$$

In triangle OAD ,

$$OA = OD = 6, \quad \angle AOD = 50^\circ$$

So

$$AD = 2(6) \sin 25^\circ = 12 \sin 25^\circ$$

The perimeter of triangle OAD is

$$12 + 12 \sin 25^\circ$$

Given that

$$\text{perimeter of sector } OBC = 2 \times \text{perimeter of triangle } OAD$$

$$(6 + x) \left(2 + \frac{5\pi}{18} \right) = 2(12 + 12 \sin 25^\circ)$$

$$x = \frac{2(12 + 12 \sin 25^\circ)}{2 + \frac{5\pi}{18}} - 6$$

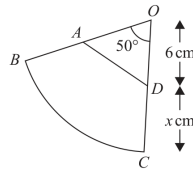
$$x = 5.8854 \dots$$

FINAL ANSWER**5.89**

PAST PAPER QUESTION**Q#: 22****Marks: 6****TOPIC: Circles, Arcs & Sectors**

Jan 2021 P2H - Jan 2021 | P2H

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Diagram **NOT**
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Angle $BOC = 50^\circ$

Given that

the perimeter of sector $OB C = 2 \times$ the perimeter of triangle OAD

find the value of x .

Give your answer correct to 3 significant figures.

$x =$

(Total for Question 22 is 6 marks)

PAST PAPER SOLUTION**Q#: 22****Marks: 6****TOPIC: Circles, Arcs & Sectors**

Jan 2021 P2H - Jan 2021 | P2H

WORKED SOLUTION*Dr. Eslam Ahmed***Topic check.** Correct.**Method**The radius of sector OBC is

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Given that

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$$(6 + x) \left(2 + \frac{5\pi}{18} \right) = 2(12 + 12 \sin 25^\circ)$$

$$x = \frac{2(12 + 12 \sin 25^\circ)}{2 + \frac{5\pi}{18}} - 6$$

$$x = 5.8854 \dots$$

FINAL ANSWER**5.89**